

Ideation Phase

Define the Problem Statements

Date	15 Feb 2026
Team ID	LTVIP2026TMIDS63806
Project Name	Heart Disease Analysis
Maximum Marks	2 Marks

Customer Problem Statement : Heart Disease Analysis

Create a Healthcare professionals, hospital administrators, and medical researchers face significant challenges in identifying meaningful patterns and risk factors related to heart disease from large volumes of raw, unstructured patient data. Without proper visual and analytical tools, understanding relationships between critical variables such as age, cholesterol levels, blood pressure, heart rate, and chest pain types becomes a time-consuming and error-prone process.

This lack of clear and immediate insight limits clinicians' ability to quickly identify high-risk patients, compare demographic trends, and make informed, data-driven decisions regarding early diagnosis, preventive strategies, and treatment planning. Static spreadsheets and raw datasets do not effectively highlight correlations, trends, or risk distributions, making it difficult to extract actionable medical insights.

There is a critical need to analyze and compare key health indicators across different patient groups — such as age categories, gender, cholesterol ranges, and blood pressure levels — to better understand how these factors influence heart disease outcomes. By developing an interactive visual analytics solution using tools like Tableau or Power BI, healthcare stakeholders can seamlessly explore trends, filter patient segments, monitor risk percentages, and identify patterns through dynamic charts and KPIs. This interactive dashboard would transform complex clinical datasets into a clear, user-friendly visual story, enabling faster diagnosis support, improved risk assessment, and more effective strategic healthcare planning in heart disease management.

Customer Problem Statement					
I am	I'm trying to	Identify	But	Because	Which makes me
Cardiologist	Identify high-risk heart disease patients	Key risk factors like cholesterol, and blood pressure	Patient data is scattered and outdated	Patient records are too complex and aged to use easily	overwhelmed
Hospital Administrator	Compare heart disease cases age & gender groups	Reports trends in heart disease by demographics	Reports are inflexible & static reports	Static, text-based files hinder segmentation	frustrated
Public Health Researcher	Understand risk factors linked to heart disease	Correlations between lifestyle factors & heart disease	Raw data doesn't highlight patterns	Unstructured data prevent clear correlation of factors	uncertain
Medical Officer	Analyze patient data to spot trends	Classify patients by risk level	Spreadsheets make trend spotting hard	Static tables hinder understanding crucial health trends	challenged
Data Analyst	Integrate large patient datasets for analysis	Raw and health trends linked to visual outcomes	Raw data is hard to transform into visual analysis	Raw data prevents clear analysis of trends & KPIs	stressed
Healthcare Planner	Develop prevention strategies	Groups at highest risk of heart disease	Difficult to generate actionable insights	Scattered unfiltered data makes locating highest risk areas	concerned

PS ID	I am	I'm trying to	But	Because	Which makes me feel
PS-1	a cardiologist	identify high-risk heart disease patients early	patient data is scattered across reports and spreadsheets	there is no centralized, interactive analytics dashboard	uncertain about diagnosis accuracy and preventive treatment planning
PS-2	a hospital administrator	compare heart disease cases across age groups, gender, and risk factors	I can't filter or analyze patient data dynamically	reports are static and not tailored to clinical decision-making	frustrated and unsure how to improve patient care strategies
PS-3	a senior healthcare executive	present heart disease trends and hospital performance insights to stakeholders	reports lack compelling visuals and clear health indicators	there is no story-driven, interactive dashboard	disengaged and less confident during strategy meetings
PS-4	a public health researcher	understand how factors like cholesterol, blood pressure, smoking, and lifestyle affect heart disease risk	I can't easily explore correlations and patterns in the dataset	existing tools are limited and non-interactive	stuck and unable to generate actionable healthcare insights